(vii) Die wiel het 'n massa M = 5 kg met R_1 = 0,1 m en R_2 = 0,5 m. Bepaal die waarde van I.

If the wheel has a mass M = 5 kg, *with* R_1 = 0,1 m *and* R_2 = 0,5 m, *determine the value of* I.

$I = \frac{1}{2}M(R_2^2 + R_1^2)$

$= \frac{1}{2}(5)(0,5^2 + 0,1^2)$

$= 0,65\ kgm^2$

2

(2)

(viii) Die wiel roteer met 'n hoeksnelheid ω = 1000 omwentelings per minuut. Bepaal die hoeksnelheid in rad/s.

The wheel is rotating with an angular velocity ω = 1000 *revolutions per minute. Determine the angular velocity in rad/s.*

1000 opm = 16,6 ops. × 2π

= 104,72 rad/s

2

(2)

(ix) Bepaal die kinetiese energie wat in die wiel gestoor is.
What is the kinetic energy stored in the wheel?

$K = \frac{1}{2}I\omega^2$

$= \frac{1}{2}(0,65)(104,7)^2$

$= 34,03\ J$

1

(2)

(x) Indien al die rotasie kinetiese energie oorgedra word aan 'n 60 W gloeilamp, hoe lank sal die gloeilamp brand?

If all the rotational kinetic energy is transferred to a 60 W globe, for how long will the lamp light up?

$P = \frac{W}{t}$

$60 = \frac{34,03}{t}$

$t = 0,567\ s$

3

(3)

8